- Open environments with complex scenes

- One model can handle all tasks

- Scalability to long videos

- Open-word vocabulary

- Each task has its own model

- Closed environments with simple scenarios

- No shared knowledge across tasks

- Closed-set vocabulary (fixed categories)

# Task-specific Learning

# Cross Task Learning

# Video Segmentation Foundation Model

VIS

# Model/Data/Loss #1

# Model/Data/Loss #2

VOS

# Model/Data/Loss #3

# PVOS

# RefVOS

# Model/Data/Loss #4

# VOS, RefVOS, PVOS, VIS, VSS, VPS

# One model / All data / All losses

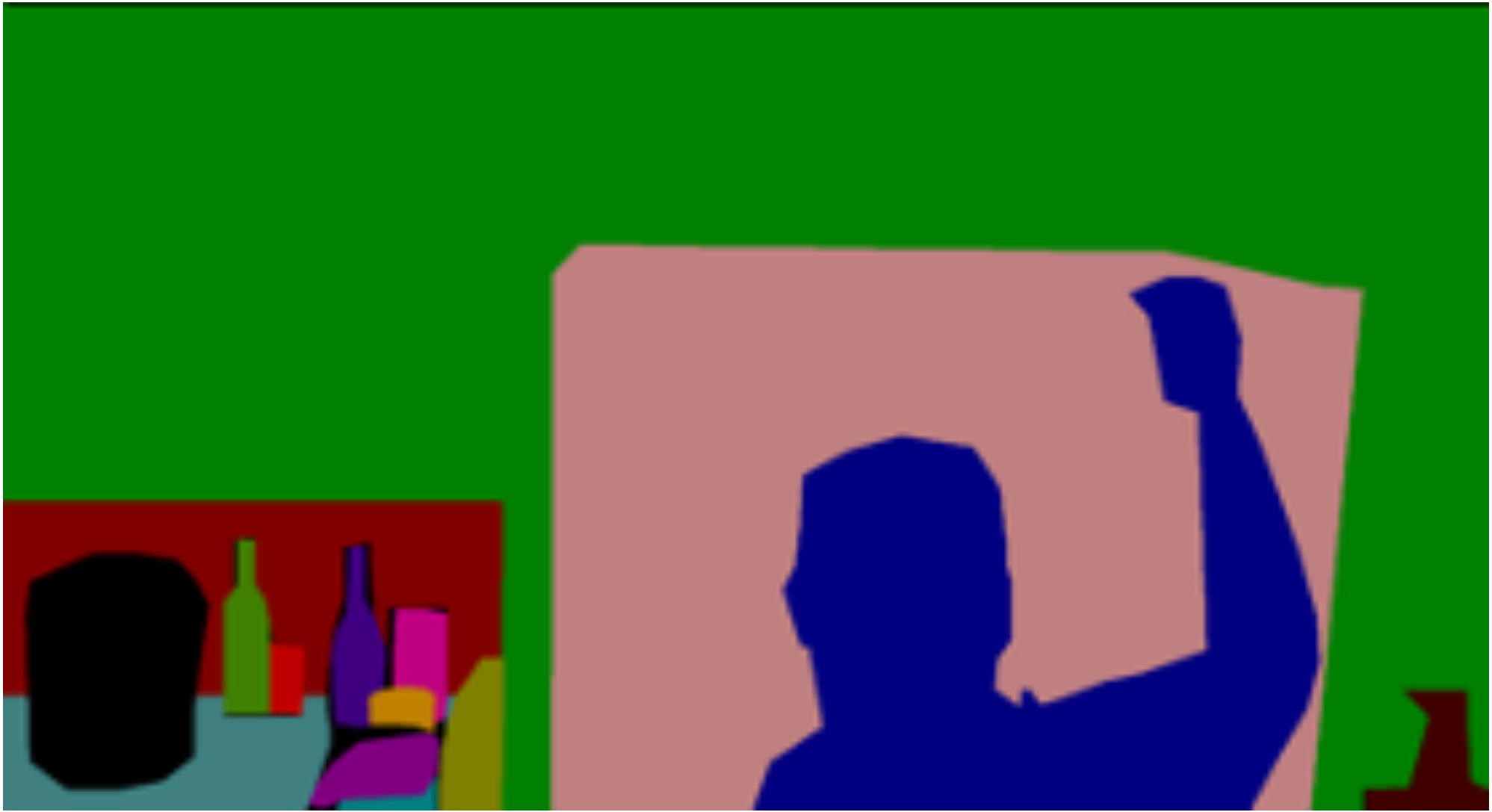
# Foundation Model

***across prompt types and time***

# *Consistent understanding*

# User #3:

# Segment all objects



# Segment the object that

# User #4:

aligned with the text prompt

***Text prompt***: A frog

in his hand and ....

is holded by a person

# User #1:

# Segment all objects

# UniVS: Video Segmentation Foundation Model

Minghan Li. **UniVS:** Unified Video Segmentation with Prompts (CVPR 2024)

# Input

# Output

# *Poor generalization*

# User #2:

# Segment the cat

# Person, Roof, Dog,

# Washer, ...

# Category names: